Mathematics Revision Test 27

1. Write the time quarter to five in the afternoon as it would appear on a 24 hour clock.

   _____

2. 1 litre costs 60p.

   Find the cost of $5\frac{1}{4}$ litres. _____

3. $\frac{1}{10}$ of a minute = _____ seconds

4. $10^2 - 9^2$ = _____

5. How much greater is 7 multiplied by 9 than 7 plus 9? _____

6. Draw a rhombus and a parallelogram showing parallel sides.

7. 0.35 x 6 = _____

8. Find how many a) cm, and b) mm, there are in 0.7m

   a) _____ b) _____

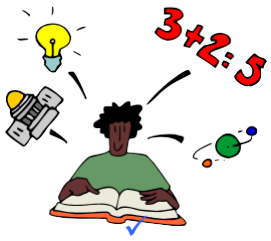
9. Find the 3 letters of the alphabet that have rotational symmetry but do not have line symmetry. _____

   Rotational symmetry means that you can turn these letters upside down and they are still the same.

10. Multiply 8.33 by 100 _____

MARKS: ____
10

Calculation Ladder



Top Tip!

Percentages are simply a type of fraction.

The symbol used for percentages is **%**.
This stands for 'per cent' which means 'out of a hundred'.

So:

50% means $\frac{50}{100} = \frac{1}{2}$

25% means $\frac{25}{100} = \frac{1}{4}$

75% means $\frac{75}{100} = \frac{3}{4}$

$33\frac{1}{3}$ % is equivalent to $\frac{1}{3}$

$66\frac{2}{3}$ % is equivalent to $\frac{2}{3}$

You must learn these off by heart so you can use them very quickly.

Ladder 5

1.	£9
2.	2 lots of this
3.	50% of this
4.	n^2
5.	Add £9
6.	Subtract £50
7.	25% of this
8.	Find 50%
9.	x 6
10.	50% of this
	Your answer is

Square Roots and Cube Roots

Revise Square Numbers and Cube Numbers before you look at this work.

SQUARE NUMBERS AND SQUARE ROOTS

The square root of 1 = 1

The square root of 4 = 2

The square root of 9 = 3

The square root of 16 = 4

The square root of 25 = 5

The square root of 36 = 6

The square root of 49 = 7

The square root of 64 = 8

The square root of 81 = 9

The square root of 100 = 10

The square root of 121 = 11

The square root of 144 = 12

This is looking at these numbers backwards.

For example:
4 squared = 16
The square root of 16 = 4

CUBE NUMBERS AND CUBE ROOTS

The cube root of 1 = 1

The cube root of 8 = 2

The cube root of 27 = 3

The cube root of 64 = 4

The cube root of 125 = 5

The cube root of 216 = 6

The cube root of 343 = 7

The cube root of 512 = 8

The cube root of 729 = 9

The cube root of 1000 = 10

The cube root of 1331 = 11

The cube root of 1728 = 12

11 PLUS PREPARATION

MATHS PAPER

Name:	
Date of Birth:	
Today's Date:	

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 35 minutes to complete the paper.
- Working out can be done on the paper or on spare paper but ensure only your answer is written clearly on the answer line.
- Each question is worth one mark. If you can't do a question, then move on.
- If you finish early then check your answers.
- Once the test has started you may not ask for help.
- No calculators should be used.

1. Answer each of the worded number questions below.

a. If one soldier eats 300g of meat a day, how many grams would 400 soldiers eat? _____

b. A soldier called Peter fired 6000 bullets through his machine gun and 500 bullets through his rifle. Another soldier called John fired 900 bullets through his rifle and 3000 bullets through his machine gun. How many more bullets did Peter fire than John? _____

c. If Peter fired 600 bullets how many would 500 soldiers fire if they did the same? _____

d. How many bullets would 300 soldiers fire if they each fired 600? _____

e. If 24,000 bullets were fired by ten soldiers in 10 days, how many would they fire in six days? _____

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2. The '?' stands for a number in the following calculations. Work out what should replace the '?' in each case.

a. 7 + ? + 6 = 23 _____

b. 45 - ? = 17 _____

c. (8+7) x (20-15) = ? _____

d. 12 x ? = 78 _____

e. (4+?) x 7 = 56 _____

---

3. Answer the following calculation questions.

a. 5.39 x 2 = _____

b. 36.08 ÷ 4 = _____

c. 16.27 – 5.34 = _____

d. 15.23 + 4.86 = _____

4. Correct the following to the nearest 100:

a.	96	= _____	**d.**	839	= _____
b.	2654	= _____	**e.**	8480	= _____
c.	13555	= _____			

---

5. Find out how the number sequence is increasing in each case and calculate the next term:

a.	2	6	12	20	30	_____
b.	6	11	16	21		_____
c.	5	7	9	11		_____
d.	2	6	14	30	62	_____
e.	1	3	6	10	15	_____

---

6. Answer the following money questions.

a. John buys three large chocolate bars for 50p each, one box of chocolates for £8.70 for his mum and one small packet of sweets for 25p. How much change will he get from £20? _____

b. Large boxes of chocolates cost £8.70. How many can be bought for £50? _____

c. By weight sweets bought in a packet cost 4/5 of those bought singly. If each single sweet costs 5p, how many sweets would there be in a packet costing £5? _____

---

7. Write each time as a fraction of an hour:

a.	15 min	_____	**d.**	35 min	_____
b.	45 min	_____	**e.**	12 min	_____
c.	6 min	_____			

8. Plot the co-ordinates shown in each question on the grid. They will reveal a shape – Name each shape.

a. (E,8) (E,10) (G,10) (G,8) _____

b. (D,3) (D,6) (E,6) (E,3) _____

c. (D,0) (F,4) (H,0) _____

d. (I,10) (J,7) (I,6) (H,7) _____

e. (J,0) (I,0) (I,3) _____

f. (A,1) (C,5) (D,5) (B,1) _____

---

9. In the following questions, A=9, B=7, C=2 and D=4. Find the solutions and write the answer in the space provided.

a. C x A x D = _____

b. B x D + A = _____

c. A + B + D = _____

d. (A x B) ÷ D = _____

1. se conversions show us how much money we would receive for £10.

£10 = US $15 **£10 = €12**

£10 = A $19 **£10 = 19 Swiss francs**

If you had £150 how much of each currency would you receive?

a. Europe _____ Euros

b. USA _____ US Dollars

c. Australia _____ Australian Dollars

d. Switzerland_____ Swiss Francs

How much would the following cost in £ Sterling?

e. A chocolate bar in Switzerland costing 19 Swiss Francs = £_____

f. A hot dog in America costing 3 US Dollars = £_____

g. A flight from Sydney to Ayres Rock costing 532 Australian Dollars = £_____

h. A bottle of wine in France costing 30 Euros = £_____

---

2. wly built office building has 64 floors of offices above ground. Below ground the building has a further ten floors. The first 3 floors below ground are storage and the next two are computer rooms. The bottom 5 floors are an underground garage.

a. the lift was on floor 12 and descended 15 floors, what floor would it be on? _____ What is on that floor? _____

b. the lift starts at floor – 10 in the car park and rises six floors, what floor would it be on? _____ What is on that floor? _____

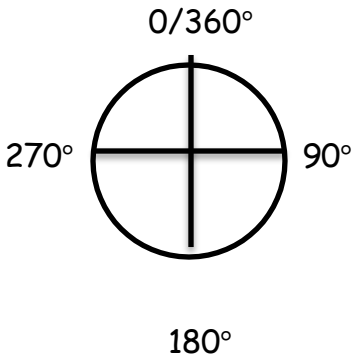
c. the lift started on floor 7 rose 42 floors then went down 36 and then down a further 15 floors, what floor would it be on? _____

d. the lift got stuck having started at -1 and then descended a further 7 floors what floor would it be on? _____

e. ift A starts at floor 57 descends 46 floors, rises 23 floors and descends 35 floors, and lift B starts at floor -8 rises 61 floors, descends 27 floors and then descends 26 floors, where would each lift end up?

Lift A _____ Lift B _____

3. diagram below shows the angles that could apply to a clock face.



a. normal clock face how many degrees is the angle between 3o'clock and 6 o'clock? _____

b. you start at 10 o'clock and turn anti-clockwise finishing at 4 o'clock, through how many degrees have you turned? _____

c. you start at 3o'clock and turn anti-clockwise finishing at 10 o'clock, through how many degrees have you turned? _____

d. you start at 12 o'clock and turn clockwise finishing at 8 o'clcok, through how many degrees have you turned? _____

e. you start at 9 o'clock and turn clockwise finishing at 6 o'clock through how many degrees have you turned? _____

f. you start at 12 o'clock and turn clockwise finishing at 5 o'clock, through how many degrees have you turned? _____

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4. wer the following worded problems:

a. tarted to paint my fence at a quarter past eleven in the morning and finished at 6.30pm. How long did it take me to paint the fence?

_____

b. mother is making an apple pie for our family of four. The recipe says to use 1.5kg of apples for six people. What weight of apples should she buy for four people? _____

5. k at the letters below and answer the symmetry questions: F

H R W

a. letters have no lines of symmetry? _____

b. ich letters have vertical symmetry? _____

c. letter has horizontal symmetry? _____

---

6. out the remainder for each calculation:

a. 46 ÷ 6 = _____

b. 37 ÷ 5 = _____

c. 126 ÷ 4 = _____

_____END OF TEST _____

If you have time

- Go back and complete any questions you missed out.

- Check your answers carefully.

CORE SKILLS 7

1

Shoppers at a supermarket were questioned about which type of bread they had in their trolley and their answers recorded in the table below:

Type of bread	No. of shoppers
White	42
Brown	28
Pitta	15
Baguette	?
Naan	13

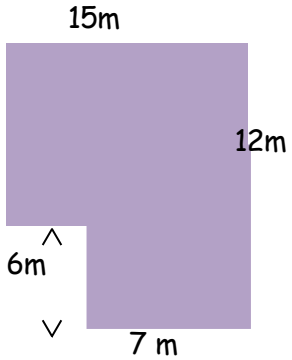
The mean number was 21, how many shoppers bought baguette?

2

Stephen spent £440 at the music shop. He spent half of the money on a drum, the rest he halved again and spent half on a violin and half on a music stand. How much money did he spend on a violin?

3

This is a plan for a new patio. What is the perimeter of the area?



4

Chicken costs £3.40 for a kilogram.
How much does 750 grams of chicken cost?

£2.30 £1.90 £2.55 £1.70 £2.80

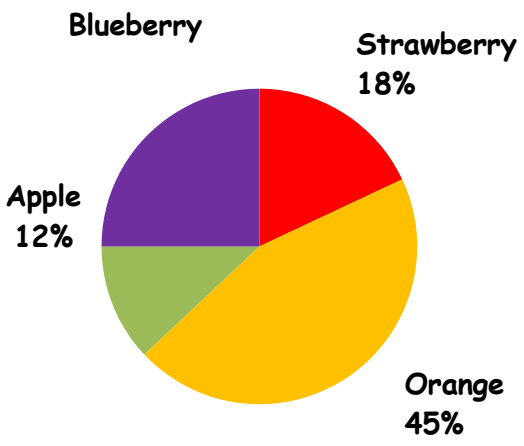
5

Simone is cooking a selection of cakes for a bake sale. She has 3 different flavours of muffin, 3 varieties of biscuit and 5 flavours of sponge. How many different combinations of muffin, biscuit and sponge can she sell?

6

In a survey 360 people were asked what their favourite fruit was. The results were recorded in a pie chart. How many people liked blueberries?

CORE SKILLS 7



7

Mark had recorded the weights of the fish he caught after a day's fishing.

2.8 Kg 4.5 Kg 880 g 8520 g 3.7 Kg 3.5 Kg 4.1 Kg

What is his median score?

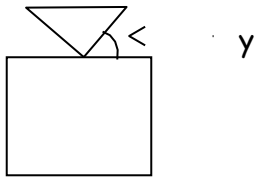
8

(Using the data in question 7)

What is his mean score?

9

In the diagram below a shape is made from a rectangle and a right-angled triangle. Choose the size of the angle y from the options below:



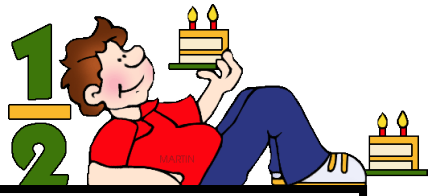
45° 90° 180° 60° 30°

10

Alice goes for a walk. She walks 2.5 km and stops for a rest. Then she walks 783 m before deciding she has taken an incorrect turn and walking back the same distance. She then continues on her journey and walks 100,000 cm. After stopping for an ice-cream she walks 1.4 km back home.

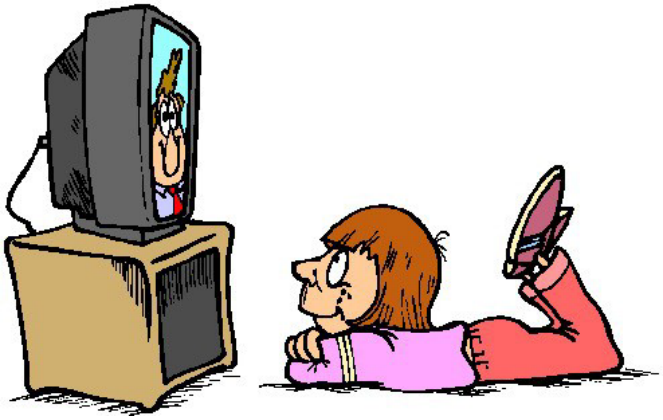
How far did Alice walk in total?

Maths Reasoning



These questions are designed to get your mind whirring - so have a go. If you can't do it first time then try, try again. It is important to persevere.

Jack just won a game show.

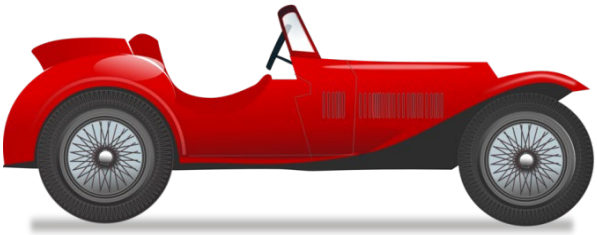


He spent one third of his winnings on a brand-new car.

He then spends two thirds of his winnings setting up a new business.

Jack is getting old so gives the £60,000 he has left to his children.

How much money did Jack win on the game show?



ANSWERS - COMPLETE MATHS – PART 29

Maths

Revision Sheet

1. 16.45

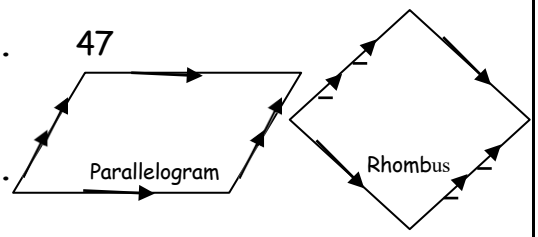
1. £3.15

2. 6 seconds

3. 19

4. 47

5.



7. 2.10

8. a) 70cm b) 700mm

9. ZSN

To find out which letters in the alphabet have Rotational Symmetry, write out the alphabet in <u>capitals</u> and ring those letters which look the same *upside down*. Try this on the lines below:

……
……
……
……
……

Have you found all 7?
N.B. H, O, I and X also have a line of symmetry which is why they are not correct answers to question 9.

10. 833

Calculation Ladder 5
Answer = £15

ANSWERS - COMPLETE MATHS – PART 29

Maths type 1 paper 4

Page 1

1.
a. 120,000
a. 2600
b.300,000
c.180,000
d. 14,400

2.
a. 10
b. 28
c. 75
d. 6.5
e. 4

3.
a. 10.78
b. 9.02
c. 10.93
d. 20.09

Page 2

1.
a. 100
b. 2700
c. 13600
d. 800
e. 8500

5.
a. 42
b. 26
c. 13
d. 126
e. 21

6.
a. 5
b. 59.55
c. 125

7.
a. 1/4 hr
b. 3/4 hr
c. 1/10 hr
d. 7/12 hr
e. 1/5 hr

Page 3

8.
a. square
b. rectangle
c. equilateral triangle
d. kite
e. Right-angled triangle
f. parallelogram

9.
a. 72
b. 37
c. 20
d. 15.75

Page 4

10.
a. 180 Euros
b. 225 US Dollars
c. 285 Australian Dollars
d. 285 Swiss Francs
e. £10
f. £2
g. £280
h. £25

11.
a. -3 storage
b. -4 computers
c. -2
d. -8
e. Lift A= -1
Lift B = ground floor

Page 5

12.
a. 90°
b. 180°
c. 150°
d. 240°
e. 270°
f. 150°

13.
a. 7hr 15 mins
b. 1kg

Page 6

14.
a. F & R
b. H & W
c. H

15.
a. 4
b. 2
c. 2

ANSWERS - COMPLETE MATHS – PART 29

Core Skills 7

1. 7
2. £110
3. 54 m
4. £2.55
5. 45 different combinations
6. 90
7. 3.7 kg
8. 4 kg
9. 45
10. 6.466 km (remember 1 km = 100000 cm)

Maths Reasoning 19

$$Total\ winnings * (\frac{1}{3} * \frac{2}{3}) = £60{,}000$$

$$Total\ winnings * (\frac{2}{9}) = £60{,}000$$

$$Total\ winnings = £60{,}000 * (\frac{9}{2})$$

$$Total\ winnings = £270{,}000$$

An alternative explanation is that from the question we know that after spending 1/3 on a new car he spends 2/3 of the remainder setting up a new business. The £60,000 he has left he leaves to his children. So we therefore know that £60,000 is 1/3 of the remainder. This means that the whole remainder after buying the car must be 3 x £60,000 = £180,000. So £180,000 was 2/3 of the original winnings. We can then work out that the original winnings must be £270,000.